Ex no: 5(b)	Managing and Operating Docker Containers: Start, Tag, Rename and File Transfers
Date:	

Aim:

The aim of this project is to understand and demonstrate the usage of Docker by managing and operating containers. This includes starting and accessing existing containers, tagging and renaming Docker images, transferring files between the host and containers, and verifying container execution. The project emphasizes optimizing containerized applications deployment, portability, and scalability.

Docker:

Docker is an open-source containerization platform that automates the process of creating, deploying, and managing containers. It enables developers to package an application along with all its dependencies into a container image, ensuring the application runs consistently across different environments. This eliminates the common problem of "it works on my machine," as the container behaves the same way whether it is running on a laptop, server, or cloud.

Purpose of Docker:

The purpose of this project is to develop practical skills in managing Docker containers and images. It focuses on starting and operating containers, tagging and renaming images, transferring files between the host and containers, and ensuring efficient deployment and portability of applications in a containerized environment.

Commands and Output:

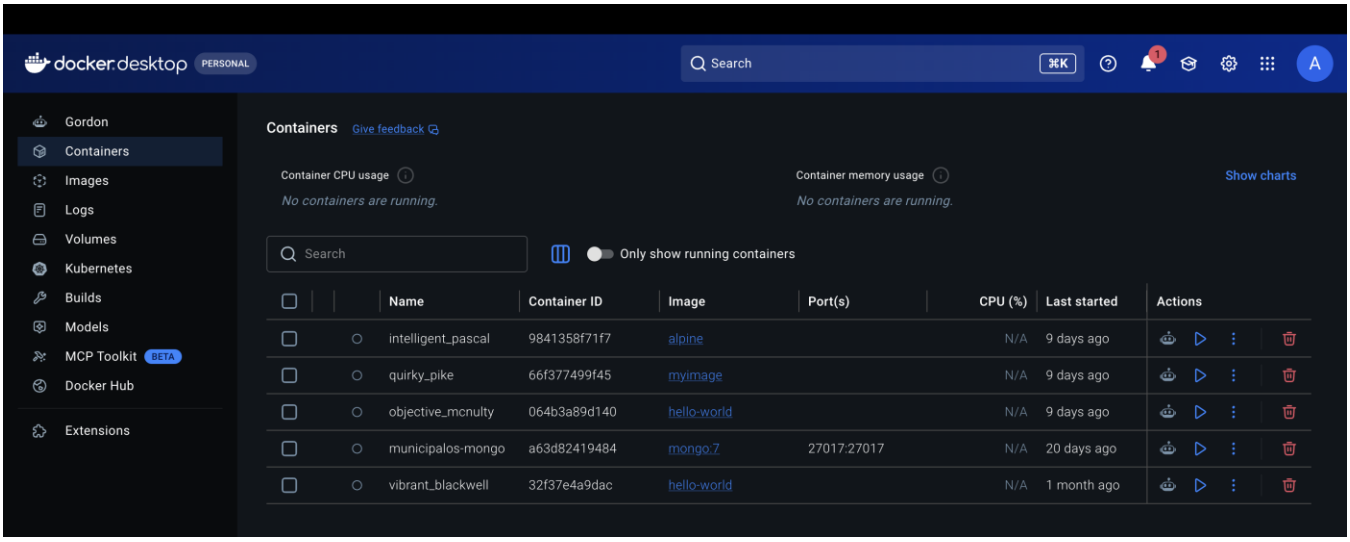
Step 1: Start Node Container

docker start <container_name>

```
C:\Program Files\Docker\Docker\resources\bin>docker start confident_hertz
confident_hertz
```

docker exec -it <container_name> node

```
C:\Program Files\Docker\Docker\resources\bin>docker exec  -it confident_hertz node
Welcome to Node.js v26.7.0.
Type ".help" for more information.
> cconsole.log("Hello");
Uncaught ReferenceError: cconsole is not defined
> console.log("Hello");
Hello
undefined
>
```



Step 2: Enter the Docker Container Using Bash

docker exec -it <docker_name> /bin/bash

```
C:\Program Files\Docker\Docker\resources\bin>docker exec -it confident_hertz /bin/bash
root@15f98f8f5df1:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@15f98f8f5df1:/# cd /usr/local/bin
root@15f98f8f5df1:/usr/local/bin# ls
docker-entrypoint.sh  node  nodejs  npm  npx
root@15f98f8f5df1:/usr/local/bin# node
Welcome to Node.js v26.7.0.
Type ".help" for more information.
>
```

Step 3: Rename the Docker Container (New cmd)

docker rename <current_container_name> <new_container_name>

```
C:\Program Files\Docker\Docker\resources\bin>docker rename node_image my_node_image

C:\Program Files\Docker\Docker\resources\bin>
```

OUTPUT:

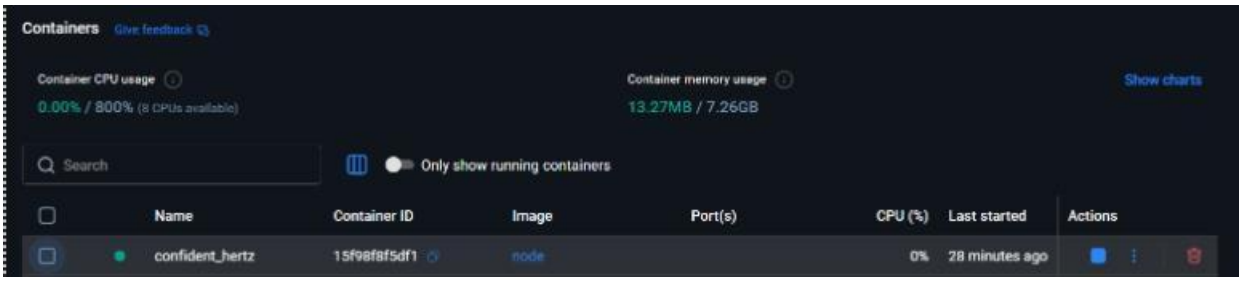


Step 4: Pull a Docker Image

docker pull alpine

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
```

OUTPUT:



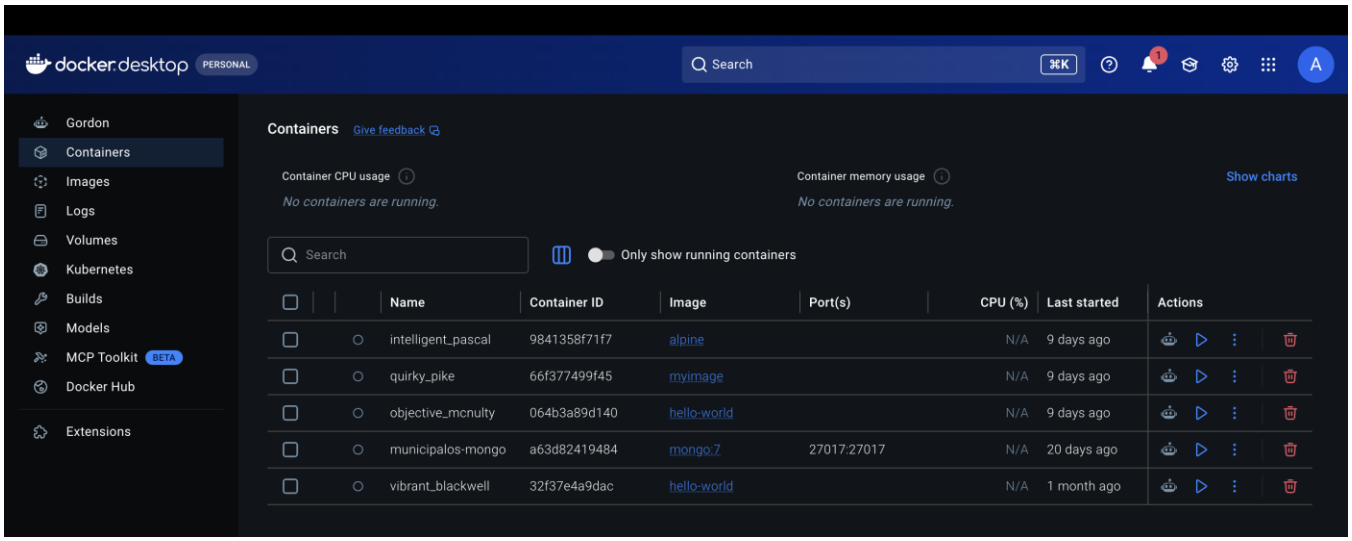
Step 5: Run Alpine Container and Explore File System

(Now go back to the previous cmd prompt and give this)

docker run -it alpine

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker run -it --name my_alpine_container alpine
/ #
```

OUTPUT:



Step 6: Copy a Local File into the Docker Container

docker cp "C:\Users\<your-path>\my-example.txt" my_alpine:/example.txt

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker cp .\my-example.txt my_alpine_container:/example.txt
Successfully copied 1.54kB to my_alpine_container:/example.txt
```

Step 7: Verify the Copied File Inside the Container

docker run -it <container_name> sh

cd /var

ls /

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker exec -it my_alpine /bin/sh
/ # ls
bin          etc          home         media        opt          root         sbin         sys          usr
dev          example.txt  lib          mnt          proc         run          srv          tmp          var
/ #
```

Step 8: Copying a File from Docker Container to Host

docker cp <container_name>:<container_path> <host_path>

```
C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile>docker cp my_alpine:/example.txt ./example_copy.txt
Successfully copied 1.54kB to C:\Users\sarat\OneDrive\Documents\Desktop\dockerfile\example_copy.txt
```

Name	Status	Date modified	Type	Size
example_copy	✓	19-08-2026 16:39	Text Document	0 KB
my-example	✓	19-08-2026 16:39	Text Document	0 KB

Problem Understanding & approach (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

The project successfully demonstrates Docker operations including container creation, image, management, file transfer, and in-container verification, validating Docker's efficiency in scalable and portable application deployment.